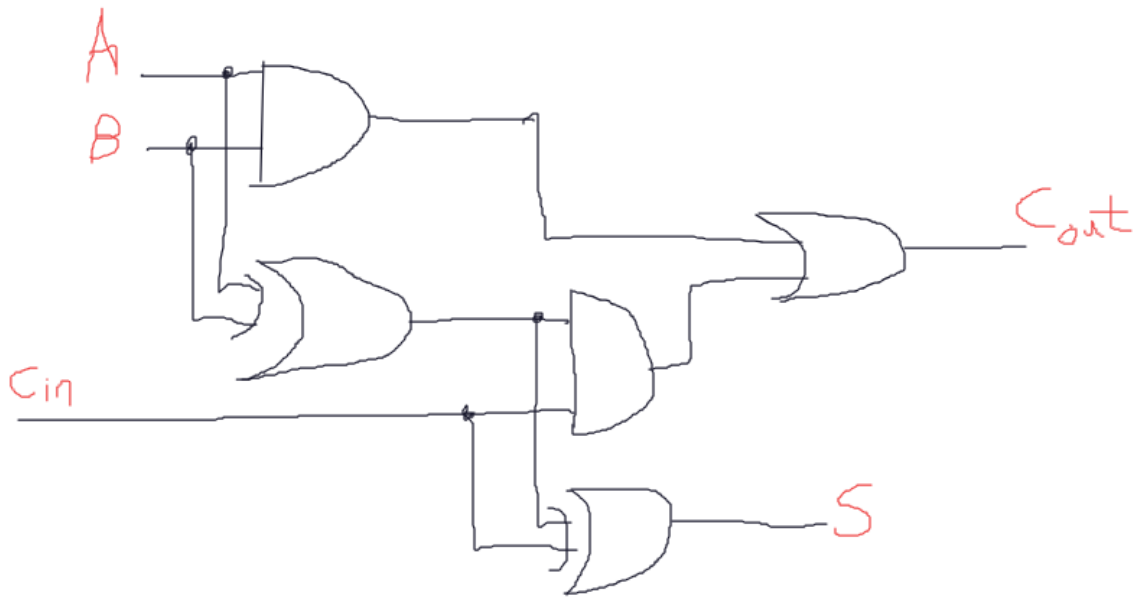




C _{in}	B	A	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

```
//Gate_level
module FA_G(
    input A, B, Cin,
    output Cout, S
);

    assign S= A ^ B ^ Cin;
    assign Cout= (A & B) | ( Cin & (A^B) );

endmodule
```